Types of gradient descent

1) batch gradient descent

2) Stochastic " "

3) mini batch " "

Variants of Gradient Descent

There are three variants of gradient descent, which differ in how much data we use to compute the gradient of the objective function.

50

```
Backpropagation Algorithm

epochs = 5
for i in range(epochs):
    for j in range(x.shape[0]):
```

5

5×5

50 = 250

Depending on the amount of data, we make a trade-off between ~~the accuracy of~~ the parameter update and the time it takes to perform an update.

trade-off: time, accuracy

- → Select 1 row (random)
- → Predict (using Forward prop)
- → Calculate loss (using Loss function → mse)
- → Update weights and bias using GD

$$W_n = W_o - \eta \frac{\partial L}{\partial W}$$

- → Calculate avg loss for the epoch

56

250 ms

Batch Gradient descent

dot Product:

y_hat = np.dot(w, x) + b

```
for i in range(nb_epochs):
    params_grad = evaluate_gradient(loss_function, data, params)
    params = params - learning_rate * params_grad
```

row = 50

epoch = 5

only 5000

for i in range(5):

batch-size → y_hat = dot-product

↳ Loss

↳ w, b update

$$w_n = w_0 - \eta \times \frac{\partial L}{\partial w}$$

batch G.D

w.u depend one.

n.e 60, 70

Stochastic G.D

50

epoch = 5 ✓ 250

5x50

→ for i in range(5):

→ shuffle

→ for j in (50): x.shape[0]

Select 1 row

→ predict

→ loss

→ weight update

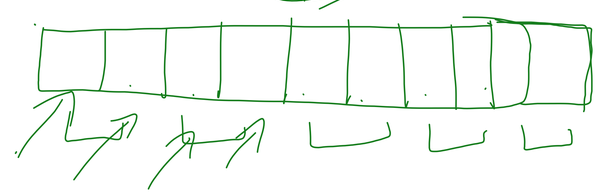
$$w_n = w_0 - \eta \frac{\partial L}{\partial w}$$

stochastic Gradient Descent is an iterative optimization technique that updates model parameters using the gradient computed from a single training example at each step.

2^n

batch size : 2, 4, 8

Keras, pytorch



row = 16

32

batch G.D : after epoch G entire dataset are 1

batch size = n

Stochastic G.D : $\boxed{\text{batch size} = 1}$

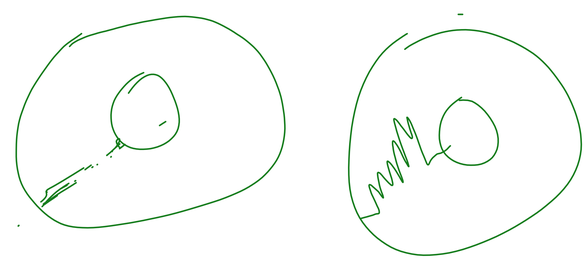
5 ⇒ to faster ?

250

1) epoch : batch faster

2) batch : $\boxed{\text{vectorization}}$

3)



Mini batch G.D:

rows = 320

batch_size = 32

32 × 16

Sol

for i in epoch:

for j in batch_size:

32

10 × 5

optimizer

→ y_pred = dot product

Loss

⇒ update

batch

Scan

mini batch

row = 320

b

epoch = 5

Batch : 5 ms

Scan : 5 × 320

= 1600

fan : 50 ms

Convergence

Sens

boxes

Mini-sa